notebooks-untitled2

May 20, 2026

1 ESCUELA POLITECNICA NACIONAL

2 FACULTAD DE INGENIERIA EN SISTEMAS

3 MÉTODOS NUMÉRICOS

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4 Taller 02 / Método de Newton vs Secante

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5 Objetivo 1

- Calcular el vector vs
- Calcular el vector f (xs) = vs

[1]:
```
import numpy as np

xs = list (range(-2, 6))
xs = list (np.linspace (-2, 6, 100))
```

[2]:
```
def f(x):
    _y = x**3 - 3 * x**2 + x - 1
    print (f"{x:.2f}, {_y:.2f}")
    return _y
```

[3]:
```
ys = [f(x) for x in xs]
```

```
-2.00, -23.00
-1.92, -21.04
-1.84, -19.19
-1.76, -17.45
-1.68, -15.83
```

```
-1.60, -14.30
-1.52, -12.88
-1.43, -11.56
-1.35, -10.33
-1.27, -9.19
-1.19, -8.15
-1.11, -7.19
-1.03, -6.31
-0.95, -5.51
-0.87, -4.79
-0.79, -4.14
-0.71, -3.56
-0.63, -3.05
-0.55, -2.60
-0.46, -2.21
-0.38, -1.88
-0.30, -1.61
-0.22, -1.38
-0.14, -1.20
-0.06, -1.07
0.02, -0.98
0.10, -0.93
0.18, -0.91
0.26, -0.93
0.34, -0.97
0.42, -1.04
0.51, -1.13
0.59, -1.24
0.67, -1.37
0.75, -1.51
0.83, -1.66
0.91, -1.82
0.99, -1.98
1.07, -2.14
1.15, -2.30
1.23, -2.45
1.31, -2.60
1.39, -2.73
1.47, -2.84
1.56, -2.94
1.64, -3.02
1.72, -3.07
1.80, -3.09
1.88, -3.08
1.96, -3.04
2.04, -2.95
2.12, -2.83
2.20, -2.67
```

```
2.28, -2.45
2.36, -2.19
2.44, -1.88
2.53, -1.50
2.61, -1.07
2.69, -0.57
2.77, -0.01
2.85, 0.62
2.93, 1.32
3.01, 2.10
3.09, 2.96
3.17, 3.90
3.25, 4.92
3.33, 6.04
3.41, 7.24
3.49, 8.54
3.58, 9.94
3.66, 11.44
3.74, 13.04
3.82, 14.75
3.90, 16.57
3.98, 18.50
4.06, 20.55
4.14, 22.72
4.22, 25.01
4.30, 27.43
4.38, 29.98
4.46, 32.66
4.55, 35.48
4.63, 38.43
4.71, 41.53
4.79, 44.77
4.87, 48.16
4.95, 51.71
5.03, 55.40
5.11, 59.26
5.19, 63.28
5.27, 67.46
5.35, 71.81
5.43, 76.33
5.52, 81.02
5.60, 85.89
5.68, 90.94
5.76, 96.17
5.84, 101.59
5.92, 107.20
6.00, 113.00
```

[4]:
```python
import matplotlib.pyplot as plt
plt.plot (xs, ys)
plt.xlabel ("xs")
plt.ylabel ("ys")
plt.title ("Grafica de ys vs xs")
plt.grid (True)
plt.show()
```

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6 Objetivo 2

- Definir la derivada de la funcion
- Leer documentación Scipy.optimize.newton
- Implementar para nuestro caso
- Verificar si esta instalado Spicy

[5]:
```python
import scipy
```

7 Alternativa 1

scipy.optimize.newton (…)

8 Alternativa 2

from scipy.optimize import newton (…)

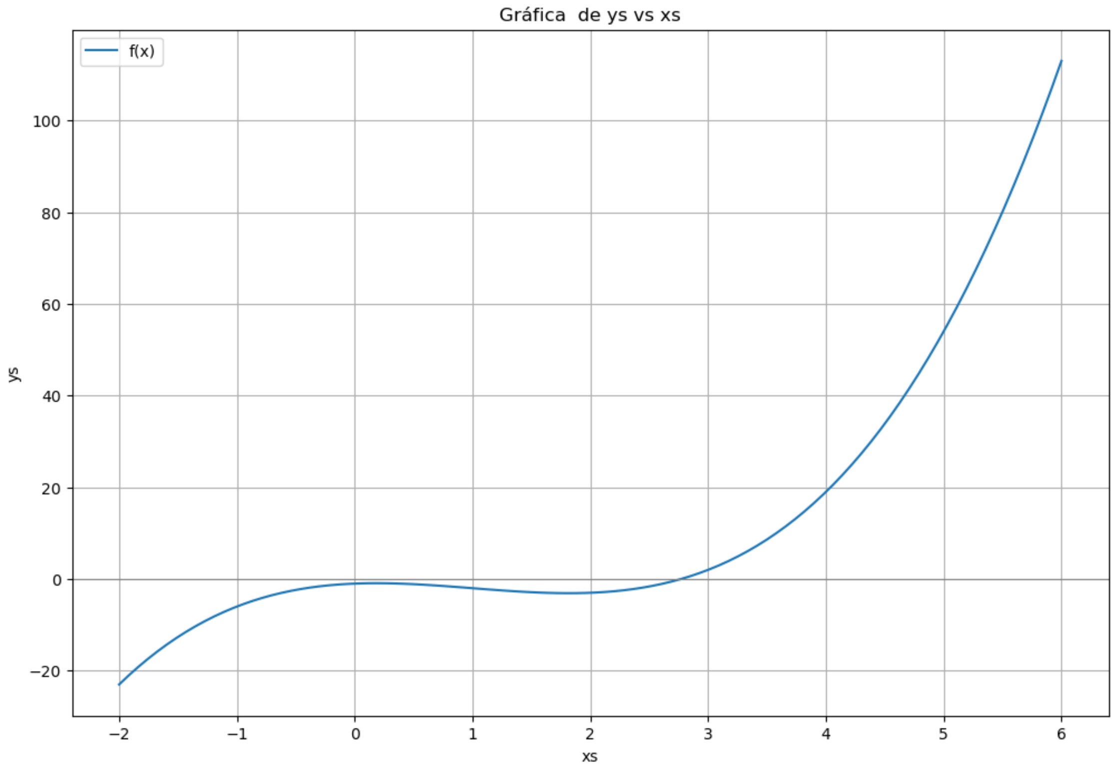
[7]:
```
from scipy.optimize import newton
newton (f, x0 = 100, fprime = lambda x: 3 * x**2 - 6 * x + 1)
```

```
100.00, 970099.00
67.00, 287421.55
45.01, 85151.64
30.35, 25223.08
20.58, 7468.63
14.08, 2209.51
9.76, 652.21
6.90, 191.43
5.03, 55.33
3.84, 15.31
3.16, 3.71
2.85, 0.59
2.77, 0.03
2.77, 0.00
2.77, 0.00
```

[7]:
```
np.float64(2.7692923542386314)
```

9 Verificamos visualmente

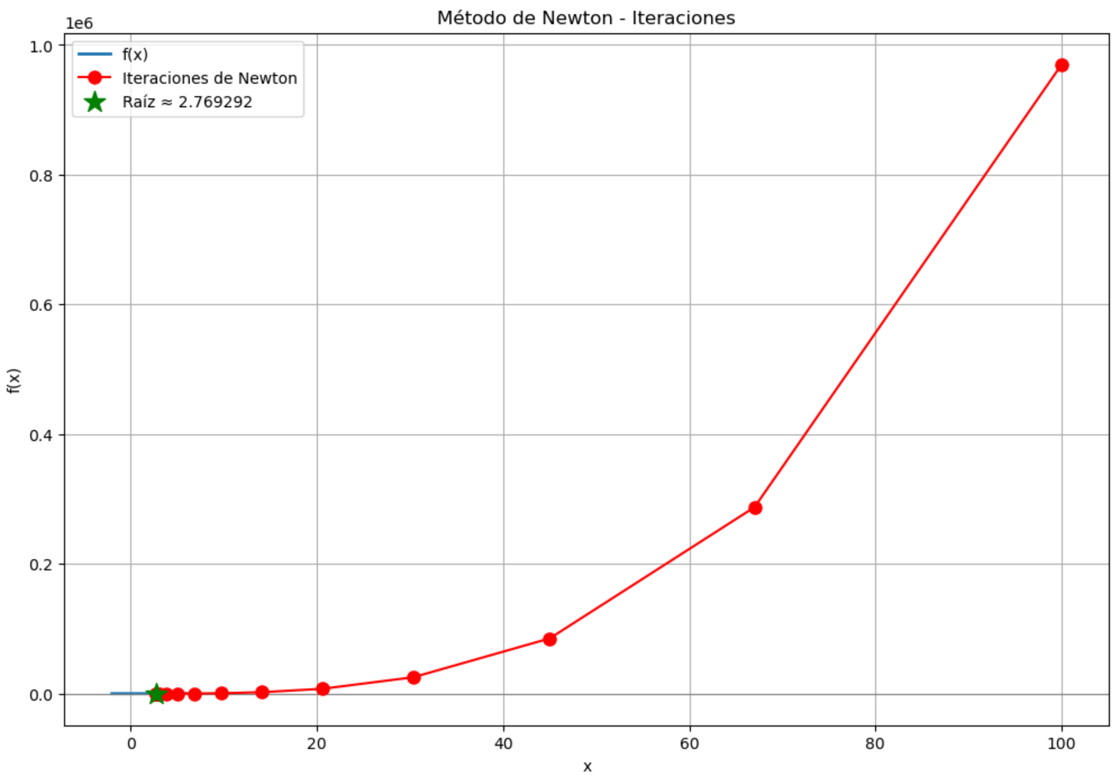
[12]:
```
# aumentar tamaño de la figura y marcar la raíz
plt.figure(figsize=(12,8))
plt.plot(xs, ys, label='f(x)')
plt.axhline(0, color='gray', linewidth=0.8)
# plt.scatter([raiz], [f(raiz)], color='red', zorder=5, label=f'raíz  {root:.
  ↪6f}')
plt.xlabel("xs")
plt.ylabel("ys")
plt.title("Gráfica  de ys vs xs")
plt.grid(True)
plt.legend()
plt.show()
```



[13]:
```
# Graficar los valores intermedios del metodo de Newton
valores_newton = []

def f_newton(x):
    _y = x**3 - 3 * x**2 + x - 1
    valores_newton.append (x)
    return _y

# Ejecutar Newton nuevamente para capturar los valores intermedios
root_nuevamente = newton (f_newton, x0 = 100, fprime = lambda x: 3 * x**2 - 6 *␣
↪x + 1)

# Graficamos
plt.figure(figsize=(12, 8))
plt.plot(xs, ys, label='f(x)', linewidth=2)
plt.axhline(0, color='gray', linewidth=0.8)
plt.plot(valores_newton, [f(x) for x in valores_newton], 'ro-', markersize=8,␣
↪label='Iteraciones de Newton', zorder=5)
plt.scatter([root_nuevamente], [f(root_nuevamente)], color='green', s=200,␣
↪marker='*', label=f'Raíz  {root_nuevamente:.6f}', zorder=6)
plt.xlabel("x")
plt.ylabel("f(x)")
```

```
plt.title("Método de Newton - Iteraciones")
plt.grid(True)
plt.legend()
plt.show()
```

```
100.00, 970099.00
67.00, 287421.55
45.01, 85151.64
30.35, 25223.08
20.58, 7468.63
14.08, 2209.51
9.76, 652.21
6.90, 191.43
5.03, 55.33
3.84, 15.31
3.16, 3.71
2.85, 0.59
2.77, 0.03
2.77, 0.00
2.77, 0.00
2.77, -0.00
```



[ ]:

```

```